Software Requirement Specification (SRS)

Farm2Market

An online Farmers Marketplace

Prepared By

Team Forerunners

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1. Introduction

Farm2Market is a web-based Farmers Marketplace designed to connect farmers directly with buyers through a simple, secure, and user-friendly online platform. The system aims to reduce dependency on middlemen, ensure fair pricing, and improve market accessibility for farmers in the digital age.

1.1 Problem Statement

Currently, many farmers rely heavily on local markets and middlemen to sell their agricultural products. This traditional system creates several inefficiencies:

- Farmers often receive lower profits due to middlemen commissions
- Farmers have limited access to buyers beyond their local areas
- There is a lack of real-time information about product demand and pricing
- Existing online marketplaces are often too complex or not designed for non-technical users
- Buyers face difficulty finding fresh produce directly from farmers

These inefficiencies reduce farmers' income, limit market transparency, and prevent fair trade opportunities. Therefore, there is a need for a simple, accessible, and secure digital marketplace tailored specifically for farmers and buyers.

1.2 Purpose

The primary purpose of Farm2Market is to develop a web-based Farmers Marketplace that connects farmers directly with buyers and ensures fair and transparent transactions.

The main objectives of the system are:

- To enable farmers to register, create, and manage their profiles easily
- To allow farmers to list agricultural products with details such as name, quantity, and price
- To provide buyers with the ability to browse, search, and place orders directly with farmers
- To implement role-based access control for Farmers, Buyers, and Admins
- To send automated email notifications upon successful order confirmation
- To provide an admin dashboard for managing users, products, and orders
- To ensure the platform is mobile-friendly and easy to use for non-technical users

Overall, the system aims to promote transparency, improve farmer's income opportunities, and create an efficient digital agricultural marketplace.

1.3 Scope

What the system will do:

- Provide user registration and secure login using email and password
- Support role-based access control
- Allow farmers to create and manage product listings
- Allow buyers to browse, search, and place orders
- Send automated email notifications upon order confirmation
- Provide an admin dashboard to manage users, products, and orders
- Ensure secure storage of user credentials
- Provide a mobile-responsive and user-friendly interface

What the system will not do:

- The system will not include online payment gateway integration in this phase
- The system will not provide logistics or delivery management
- The system will not include advanced analytics or demand forecasting features

2. Design and Implementation Constraints

2.1. Programming Language

The system will be developed using:

- **Backend & Frontend:** Python (Django Framework)
- **Frontend Rendering:** Django Templates (HTML, CSS, Tailwind)

Justification:

- **Python** is selected for its simplicity, readability, and strong community support.
- **Django** is chosen because it provides:
  - Built-in authentication system
  - Secure password handling
  - Admin panel support
  - ORM for database management
  - Rapid development features

- **Django Templates are used for the frontend to:**

- Enable structured and efficient development using Django's built-in template engine
- Render dynamic content directly from the server
- Ensure seamless integration with backend logic and database models
- Simplify maintenance within a unified system architecture

2.2 Database & Servers

- **Database**: MySQL (MySQL ready)
- **Architecture**: Monolithic Django Web Application

Data Storage & Retrieval:

- All user data, product listings, and orders will be stored in a relational database
- Django ORM will handle all database operations securely
- The system will follow an MVC (Model-View-Template) architecture:
  - Models → Database structure (Users, Products, Orders)
  - Views → Business logic and request handling
  - Templates → Frontend UI rendering

3. User Classes and Characteristics

There are different categories of users in the Farm2Market system, each being assigned a role, responsibility, and permission. The following is a summary of each category in my opinion, as a learner who was studying this system.

3.1. Guest User

Description:
Unregistered users who visit the platform without creating an account.

Technical Expertise:
- Low to Medium
- General knowledge of using smartphones.

Permissions:
- Browse products
- Search product
- Register an account
- Login

Characteristics:

- Accountable for system integrity.
- Check that the platform works well.
- Maintain a high level of security and adherence to policies by users.

3.2. Farmers

Description:
The major users are farmers who make their agricultural products available in the site.

Technical Expertise:
- General knowledge of using smartphones.
- Low level of technical expertise.
- Like a minimal and non-complex interface to avoid being stuck.

Permissions:
- Log in and register for the system.
- Manage and create their profile.
- Add, edi,t and remove product listings.
- See orders made on their products.
- Stock product availability.
- Email notifications of orders.

Characteristics:

- Focused on selling products
- Access via mobile is easy because it is convenient.
- Must have an easy and user-friendly interface that is not cluttered.

3.3. Buyers

Description:
The buyers may be either individuals, retailers, or businesses that wish to buy goods straight from the farmers

Technical Expertise:
- Online shopping and basic internet knowledge.
- Browsing and making orders with ease.

Permissions:
- Register and log in
- Browse and search products
- Add products to cart
- Place and confirm orders
- View order history
- Get notified via email.

Characteristics:

- Interested in new commodities.
- Look forward to a speedy and hassle-free ordering process, which suits their hectic schedules.

3.4. System Administrators

Description:
Admins manage and watch the work of the whole system to ensure that everything is going alright.

Technical Expertise:
- Intermediate and high technical expertise.
- Knowledgeable of web-based dashboard and administration tools.

Permissions:
- Access the admin dashboard
- View and manage users
- Monitor product listings
- Monitor and manage orders
- Deactivate or lock out accounts when necessary.

Characteristics:

- Accountable for system integrity.
- Check that the platform works well.
- Maintain a high level of security and adherence to policies by users.

4. Functional Requirements

ID	Requirement Name	Description	Users	Priority
FR-1	User Registration	The system shall allow users to register as either a Farmer or Buyer.	Guest	High
FR-2	User Login & Logout	The system shall allow registered users to securely log in and log out.	Buyer Farmer Admin	High
FR-3	Profile Management	The system shall allow users to view and update their profile information.	Buyer Farmer	Medium
FR-4	Add Product	The system shall allow farmers to add new agricultural products.	Farmer	High
FR-5	Update Product	The system shall allow farmers to edit their listed products (stock availability, price update).	Farmer	High
FR-6	Delete Product	The system shall allow farmers to remove their listed products.	Farmer	High
FR-7	View Orders	The system shall allow farmers to view orders placed for their products.	Farmer	High
FR-8	Browse Products	The system shall allow buyers and guests to browse available products.	Guest Buyer	High
FR-9	Search Products	The system shall allow users to search products by name or category.	Guest Buyer	High
FR-10	Add to Cart	The system shall allow buyers to add products to a shopping cart.	Buyer	High
FR-11	Update Cart	The system shall allow buyers to modify cart items.	Buyer	High
FR-12	Place Order	The system shall allow buyers to place an order.	Buyer	High

FR-13	Payment Processing	The system shall allow buyers to select and complete a payment method (Cash on Delivery or Online Payment).	Buyer	High
FR-14	Order Status Tracking	he system shall allow users to track order status. (Pending, Confirmed, Shipped, Delivered).	Buyer Farmer Admin	High
FR-15	Product Review	The system shall allow buyers to rate and review purchased products.	Buyer	Medium
FR-16	Admin Features	The system shall allow the admin to manage registered users.	Admin	High
FR-17	Manage Categories	The system shall allow the admin to manage product categories.	Admin	High
FR-18	Monitor Transactions	The system shall allow the admin to monitor orders and transactions	Admin	High

5. Non-Functional Requirements

All these requirements are concerning the ease and security of the Farm2Market system. They include the performance, security, ease of use, reliability, and ease of maintaining it in the long term.

- **Performance Requirements:**

  Response Time:
  It should take less than 3 seconds to respond when you log in, browse through products, or place an order, provided everything is functioning properly with the system.

  Concurrent Users:
  Our objective is to have 100 or more individuals utilizing the app simultaneously without it becoming sluggish over time as our university project becomes popular.

  Data Processing:
  The listings of products, search results, and confirmation of orders have to load practically instantly, leaving you without waiting.

  Notification Delivery:
  The email confirmation of the order must appear within a minute of clicking the submit button

- **Security Requirements:**

  Authentication Security:
  The passwords will be saved in a secure place using the built-in hash system of Django, and nobody will be able to access them in their plain-text form.

  Role-Based Access Control:
  This is because different users (Farmer, Buyer, Administrator) can only see what they are meant to see, and no unauthorized snooping is allowed.

  Data Protection:
  Sensitive information such as passwords and contact information will not be left sitting in the open or tossed into the APIs in plaintext.

  Input Validation:
  Any entry made by a user is verified to prevent typical malpractices such as SQL injection and XSS to make the platform responsible.

  Secure Communication:
  We shall use HTTPS just in order to ensure that all data remains confidential during transit.

- **Usability Requirements:**

User-Friendly Interface:
The interface will be straightforward, user-friendly, and mobile-friendly; which is exactly what you would want in case you are busy with classes and other activities.

Browser Compatibility:
It can also be used on any device since it works cleanly with the latest browsers and even with standard smartphones.

- **Reliability Requirements:**

System Availability:

We aim to achieve a minimum of 95 per cent uptime through the project, with no consideration of planned maintenance, which will be no secret to all.

Error Handling:

There will also be clear and helpful error messages to indicate that something has gone wrong, rather than the app simply crashing.

- **Maintainability Requirements**

Modular Design:

The interface will be straightforward, user-friendly, and mobile-friendly; which is exactly what you would want in case you are busy with classes and other activities.

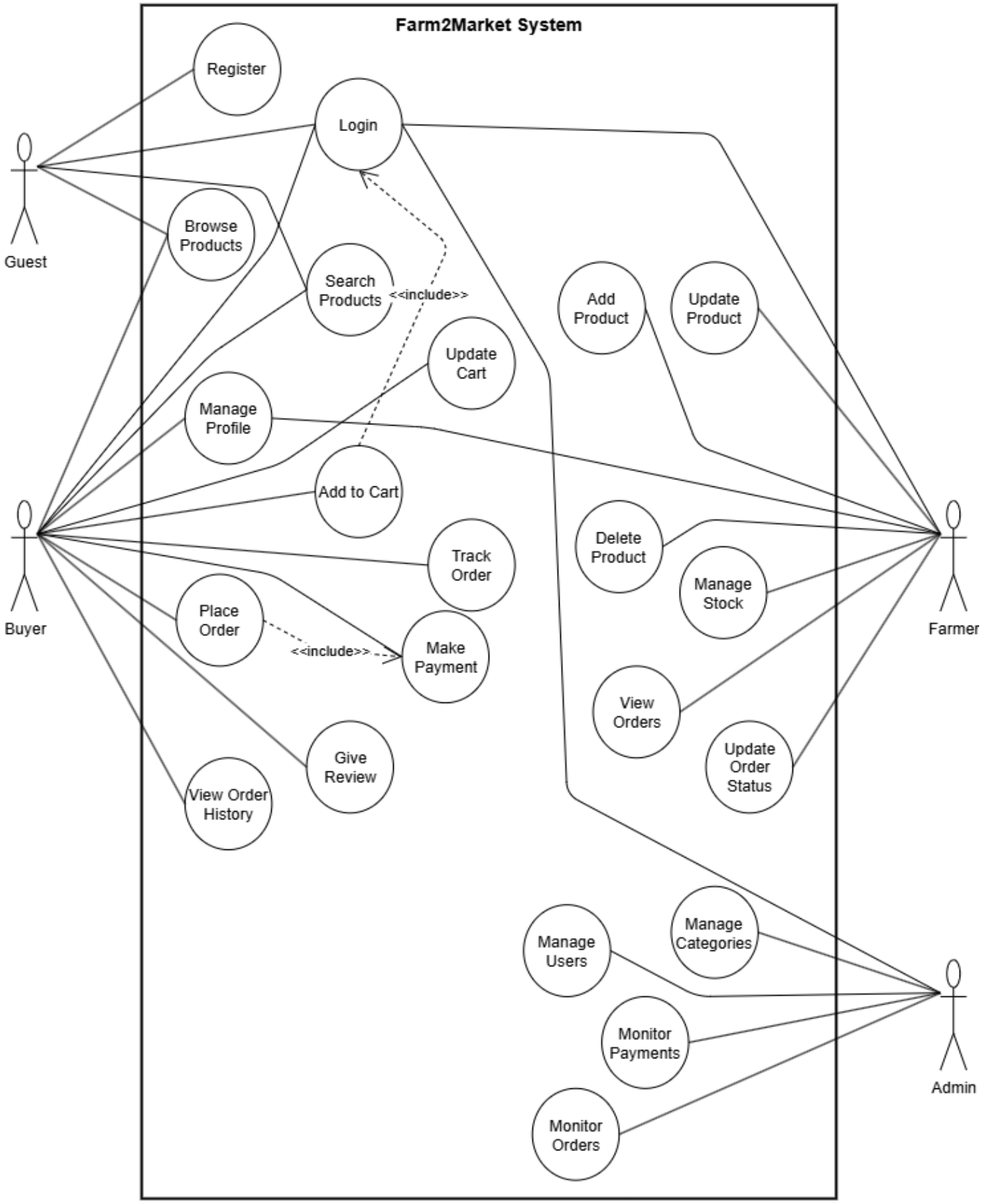
Documentation:

There is good technical documentation and user manuals, and thus maintaining the project will not be a burden.

6. System Design Diagrams

Visual representations are required to understand system logic.

6.1.Use Case Diagram



6.2 Use Case Descriptions

1. Use Case: User Registration
Actor: Guest (Future Farmer or Buyer)
Process: When a new user creates an account with the platform to gain access to role-specific functions.
Precondition: The guest clicks to the Farm2Market registration screen.
Main Success Flow:
The guest chooses the alternative of creating a new account.
The system will ask a user to enter his or her registration information, which includes his or her email address and role of choice (Farmer or Buyer).
The guest fills in the necessary details and fills in the form.
The information given is verified by the system.
The system is effective in creating the new user profile and storing the credentials safely.
The system provides a message of success and directs the user to the login page.

2. Use Case: Add Product Listing
Actor: Farmer
Description: A farmer prepares a new item of agricultural product listing to be seen by the buyers.
Precondition: The farmer is logged in the system with their email, password.
Main Success Flow:
The farmer clicks on to their profile or dashboard interface.
The farmer uses the option of adding a product.
The system has a clear product listing form.
The producer types the product information, namely the name of the product, its quantity, and fair price.
The listing of the products is provided by the farmer.
The system authenticates the input information and writes the new listing in the database.
The system shows the active product list of the farmer and opens the item to the buyers on the marketplace.

3. Use Case: Browse and Search Product
Actor: Buyer
Description A buyer searches on the fresh produce or on the agricultural products available.
Precondition: The buyer has entered the platform (may be logged in or surfing as a guest, according to your diagram).
Main Success Flow:
The purchaser is directed to the front office market.
The system shows a list of the existing products that are offered by different farmers.
The customer types certain keywords in the search box or can choose a section.
The system works with the search query and filters the database.
The system searches and presents a narrowed down list of compatible products to the buyer.

4. Use Case: Place Order

Actor: Buyer

Description: A buyer verifies that commodities that he/she has chosen in a farmer have been purchased.

Precondition: The buyer is logged in safely and added products wanted to his/her cart.

Main Success Flow:

The customer will go to his/her cart and check out the products.

The buyer starts the checkout process by clicking on the button of Place Order.

The final order summary is displayed in the system.

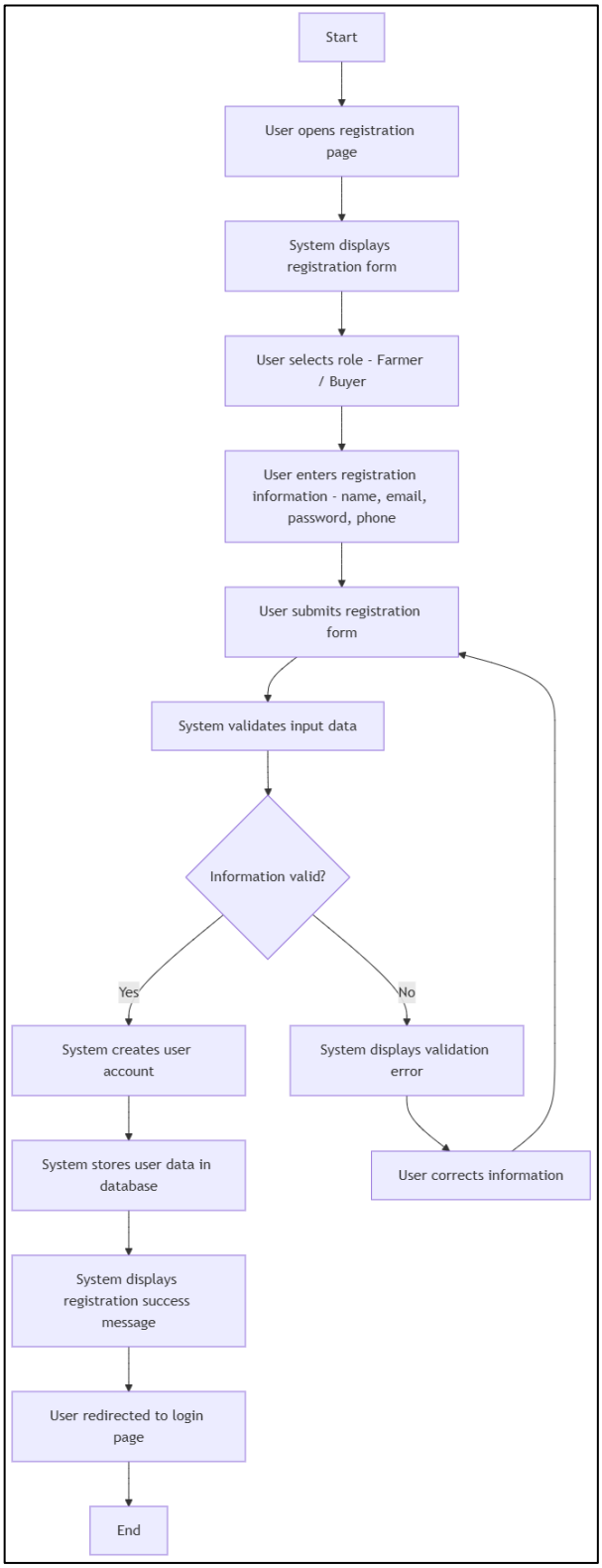
The buyer acknowledges order placement.

The system is able to make the order recorded in the database and connected to the corresponding farmer.

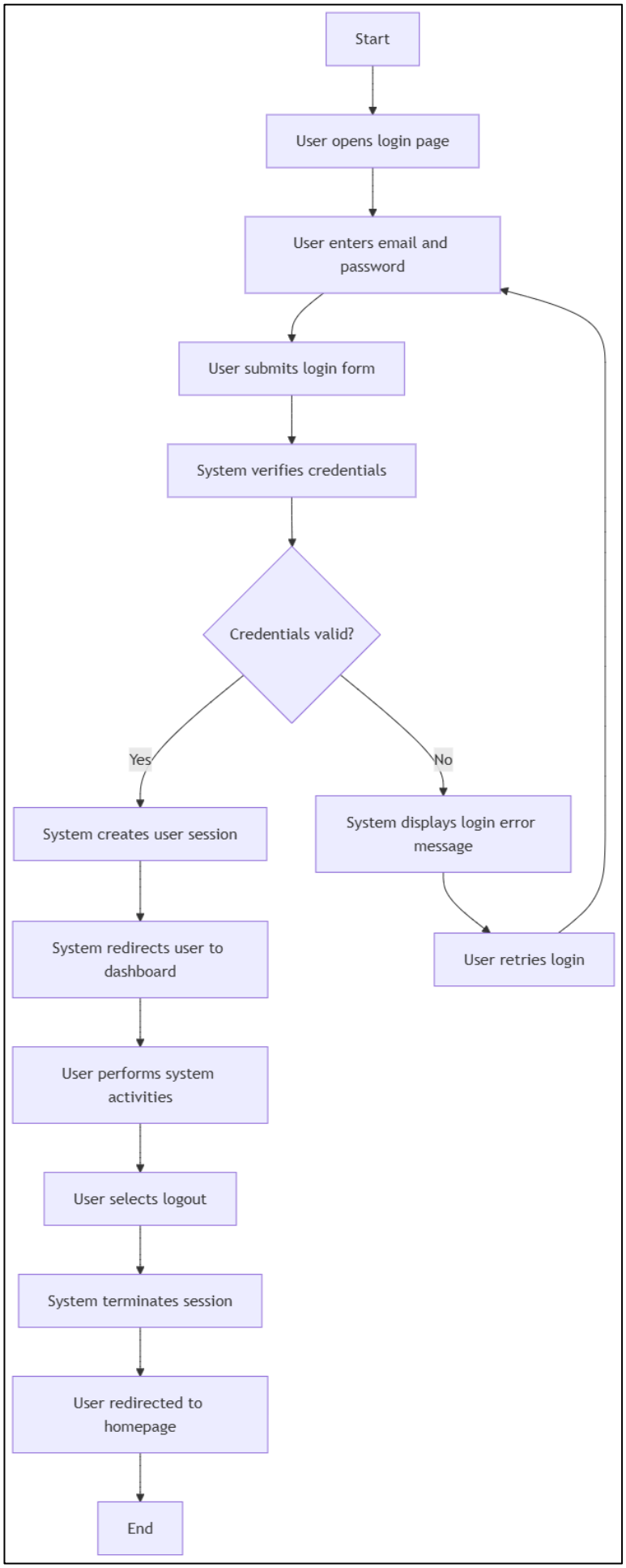
The system will send an automated email notification that will confirm the details of the order and will be sent to the buyer.

6.3 Activity Diagram

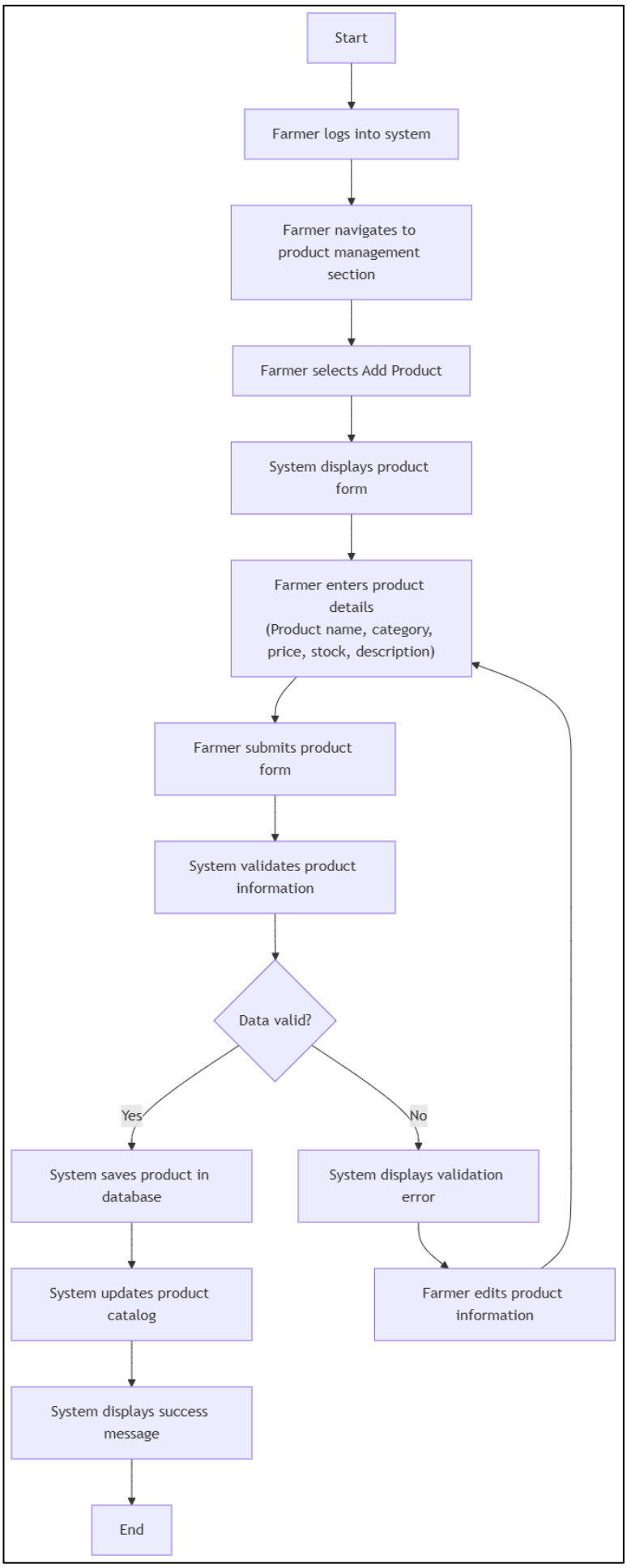
Registration:



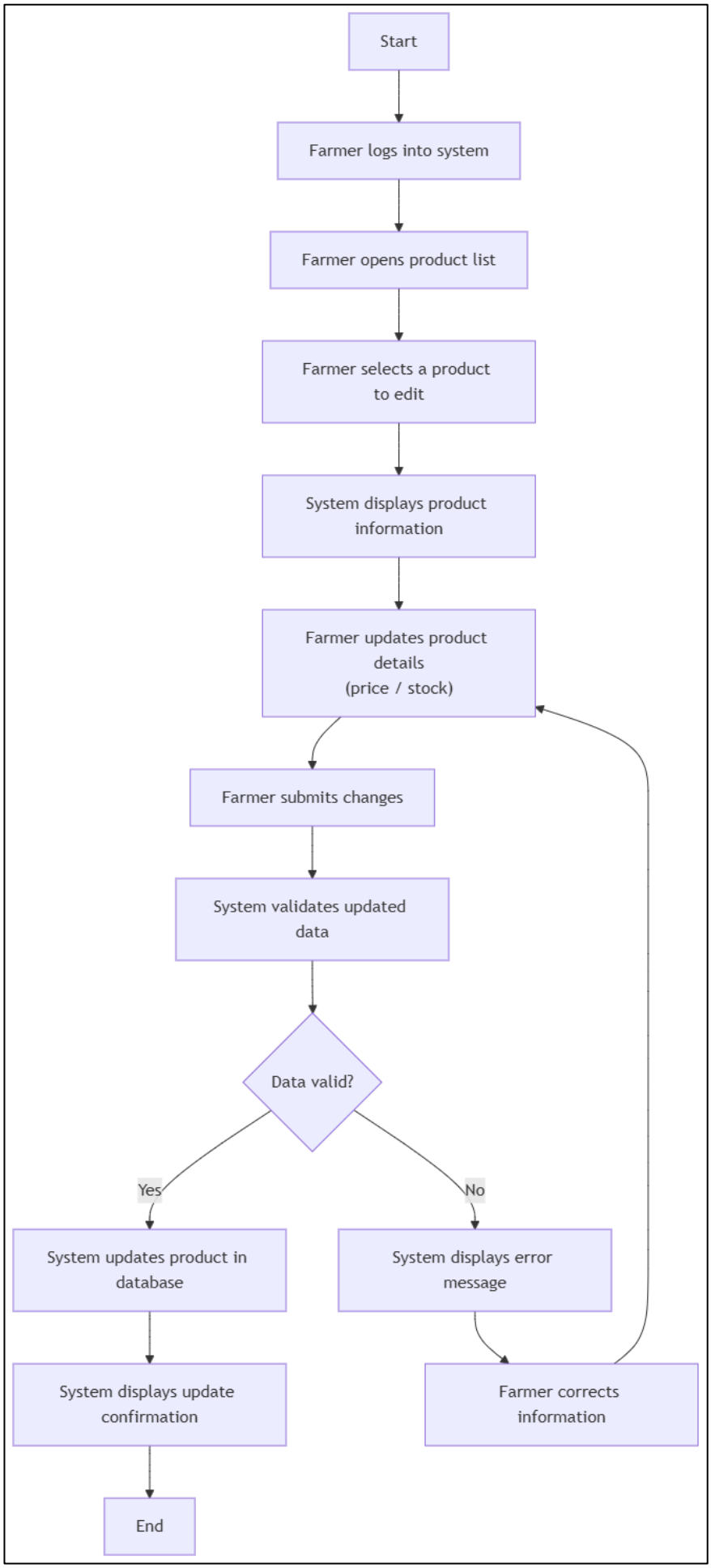
Login:


Start
User opens login page
User enters email and password
User submits login form
System verifies credentials
Credentials valid?
Yes
No
System creates user session
System displays login error message
System redirects user to dashboard
User retries login
User performs system activities
User selects logout
System terminates session
User redirected to homepage
End

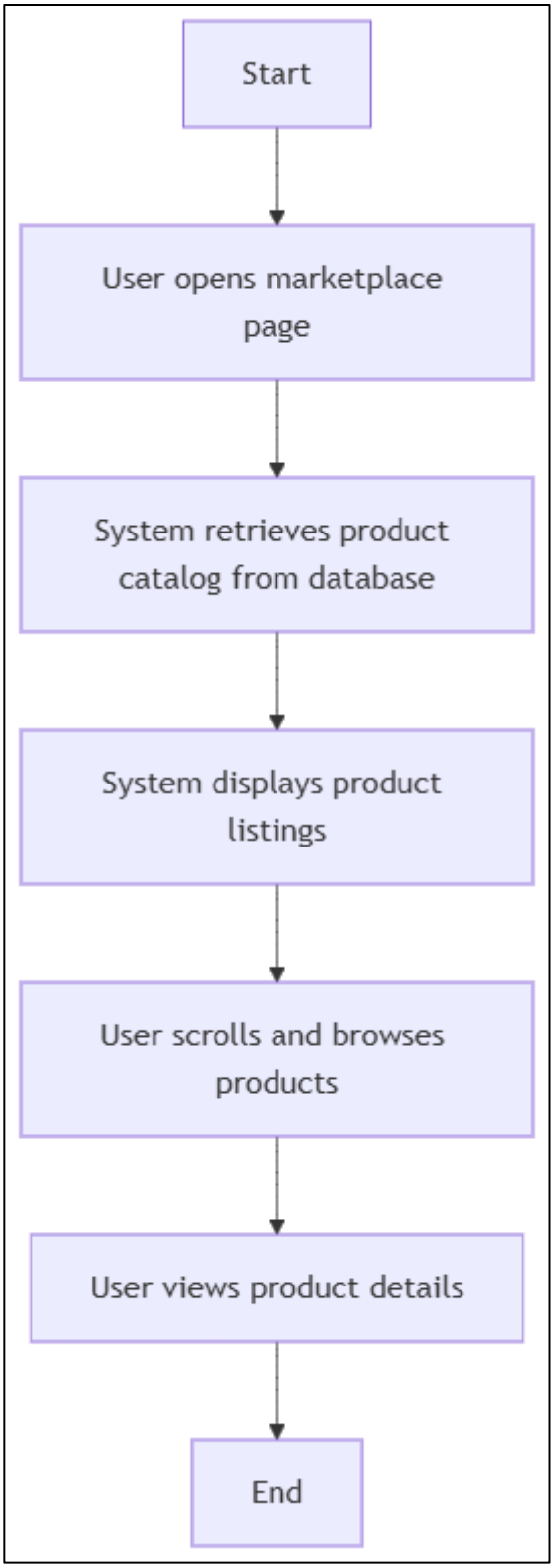
Add Product:


Start
Farmer logs into system
Farmer navigates to product management section
Farmer selects Add Product
System displays product form
Farmer enters product details (Product name, category, price, stock, description)
Farmer submits product form
System validates product information
Data valid?
Yes
No
System saves product in database
System displays validation error
System updates product catalog
Farmer edits product information
System displays success message
End

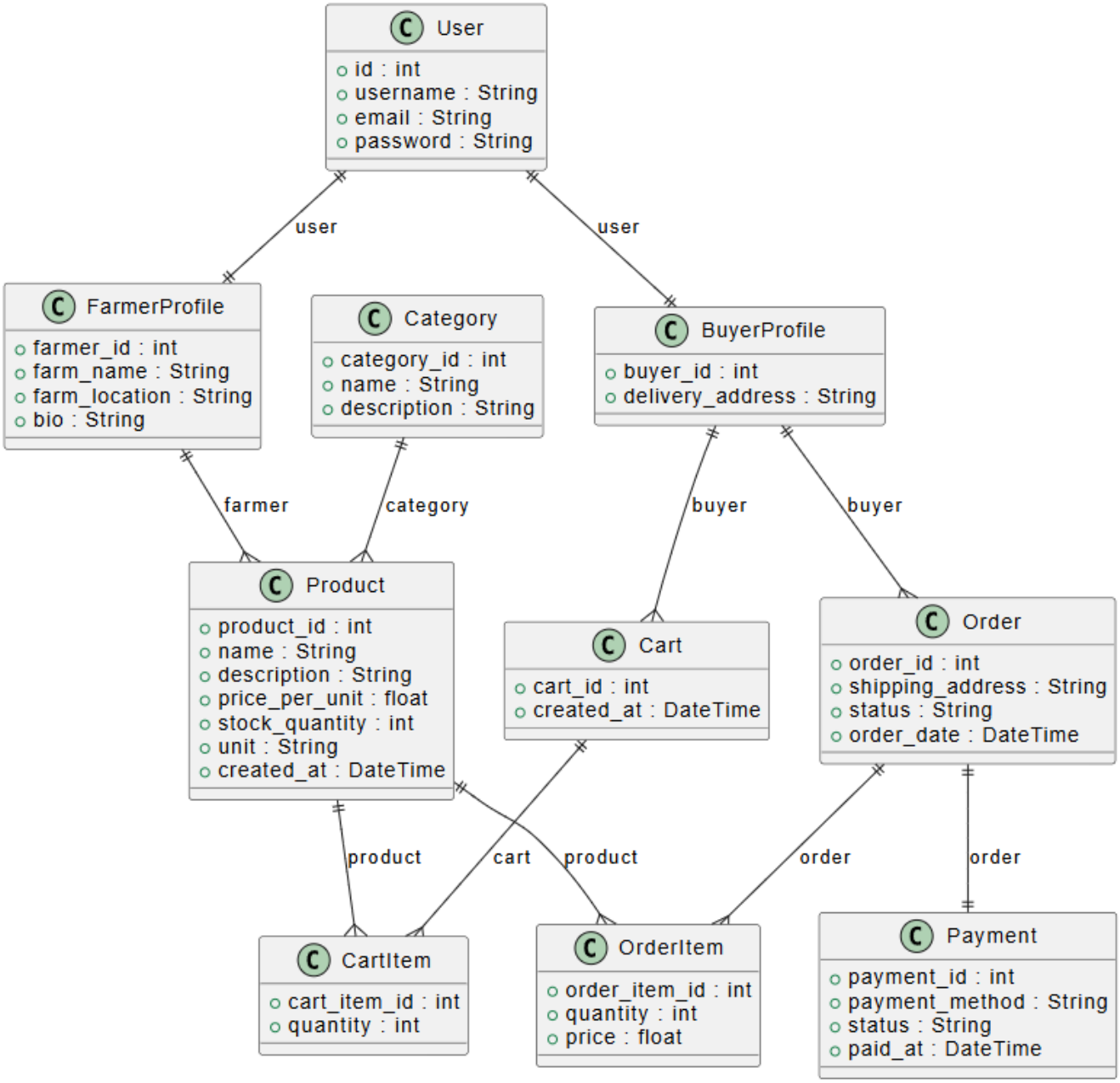
Update Product:



Browse Products:



6.4 Entity Relationship Diagram (ERD)



7. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Chowdhury Nabil Ahmed	0242310005341011	Backend, Frontend
MD Shardul Rahman Turjo	0242310005341263	Backend, Frontend
MD Sajib Ahammad	0242310005341473	Database, ORM, Documentation,